Concepts of Computational Geometry and its Applications in Robotics

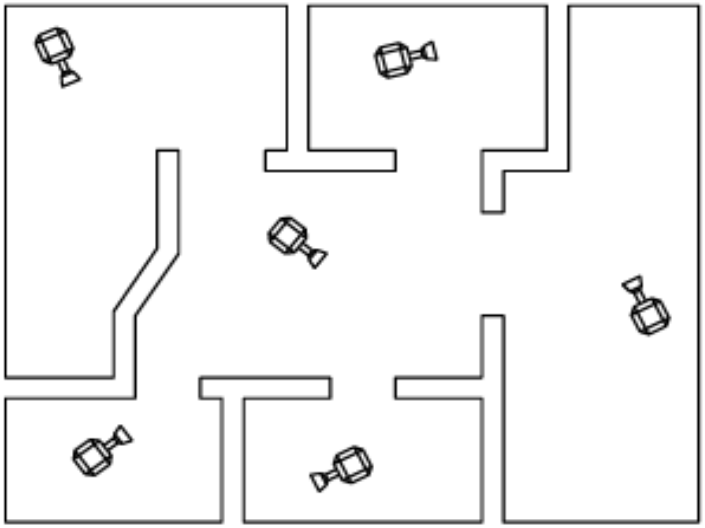
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Polygon Triangulation

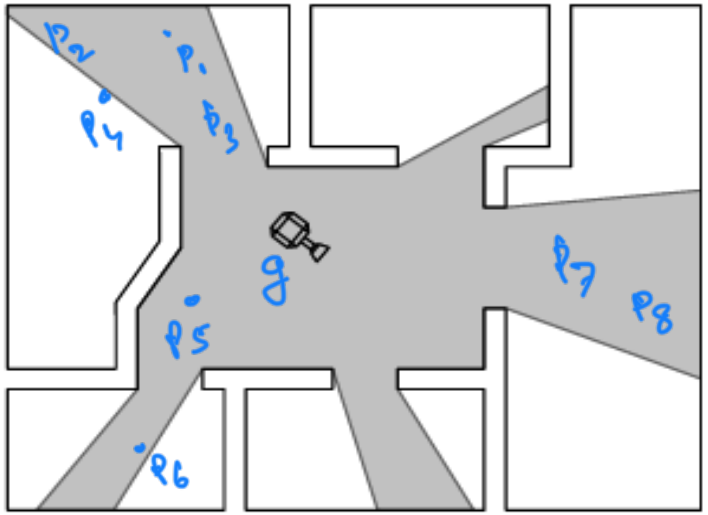
Application – Guarding an Art Gallery

- The Art Gallery Problem is a classic problem in computational geometry.
- Given a simple polygon with n vertices, what is the minimum number of stationary guards (cameras) sufficient to see every point inside the polygon?



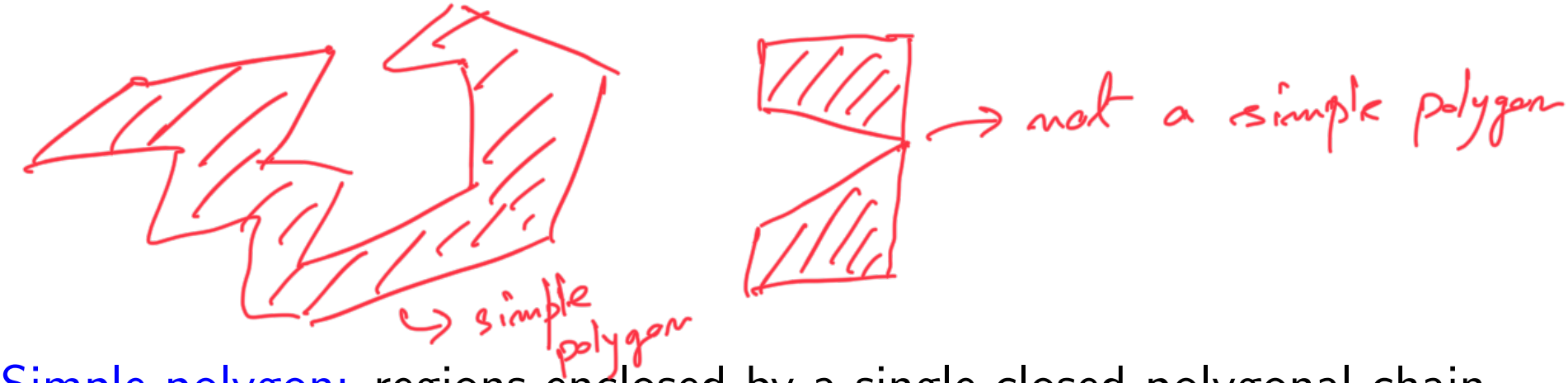
Application – Guarding an Art Gallery

- A camera sees those points in the polygon to which it can be connected with an open segment that lies in the interior of the polygon.
- A guard at a point g can see a point p if the entire line segment $\overline{gp}$ lies inside or on the boundary of the polygon.



Difference with CSPP is to be understood.

Art Gallery Problem



- Simple polygon: regions enclosed by a single closed polygonal chain that does not intersect itself.
- How many cameras do we need to guard a simple polygon?

Art Gallery Problem

Guarding simple polygons

- Depends on the polygon at hand
  - the more complex → the more no. of cameras.
- Is there any bound on the number of cameras needed?
  in terms of n (the no. of vertices of the polygon)
- Even when two polygons have the same no. of vertices, the no. of cameras may differ.



1



Convex polygons.

c1:







c2:





Non-convex

Art Gallery Problem

Worst-case scenario:

- Bound that is good for any simple polygon with n-vertices

- Convex polygons are easy to guard
  - Simplest convex polygon: Triangles

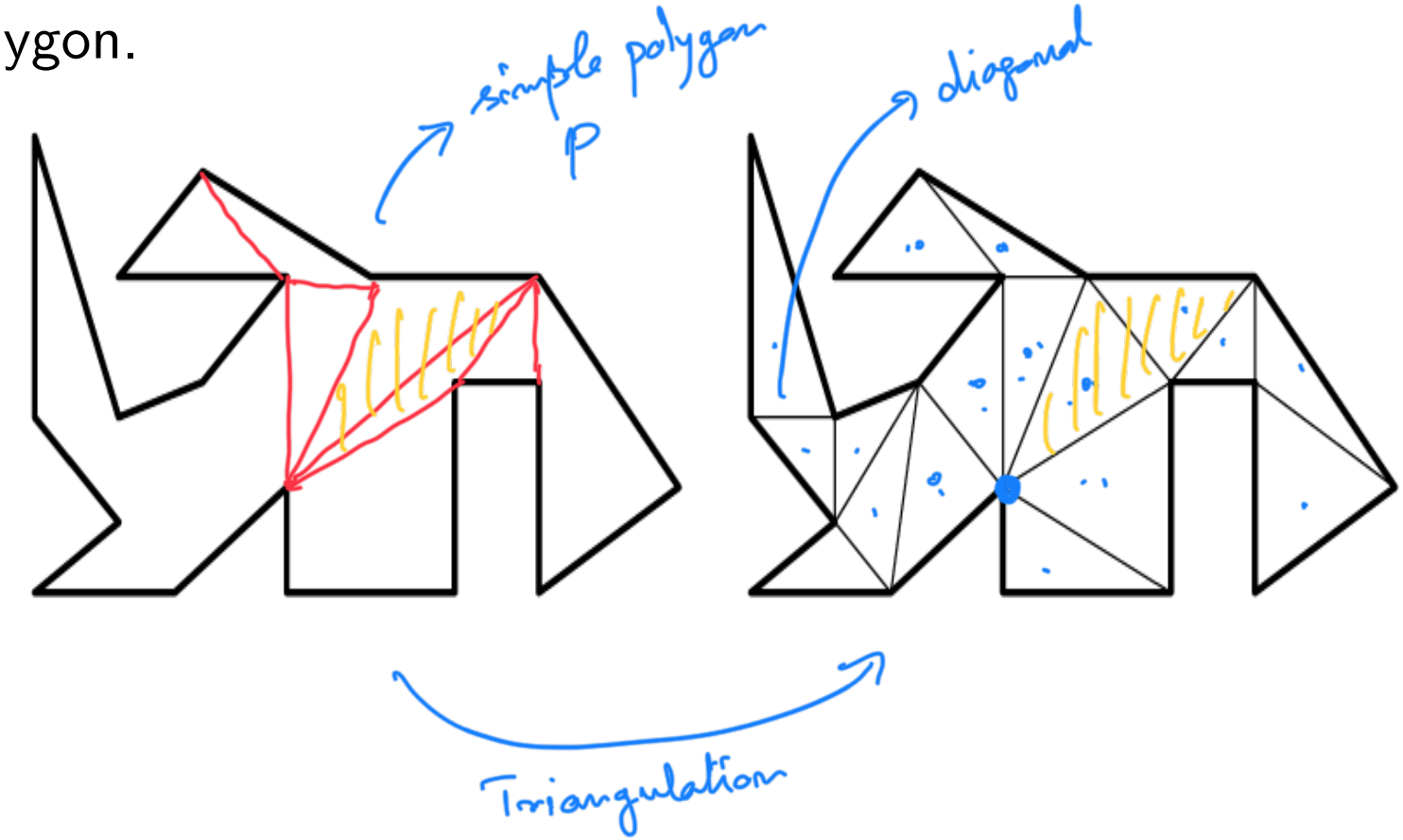
Why don't we directly solve it for a given simple polygon?

- Algorithmically it is difficult / intractable to solve. (NP-hard)

Art Gallery Problem – Triangulation

P: a simple polygon with n vertices

- Triangulation: A decomposition of a polygon into triangles by a maximal set of non-intersecting diagonals is called a triangulation of the polygon.



Art Gallery Problem

(i) Is triangulation of a given simple polygon unique? NO

(ii) Does a triangulation exist? YES

(iii) How many triangulation can there be?

Art Gallery Problem

Why $\lfloor n/3 \rfloor$ cameras suffice?

1. Triangulate the polygon

   n vertices $\longrightarrow$ $n-2$ triangles

2. 3-coloring of the vertices

3. Put cameras at all vertices of the least used color.

$\lfloor n/3 \rfloor$

n_R, n_G, n_B

$n_R + n_G + n_B = n$

$\min\{n_R, n_G, n_B\} \Rightarrow$ Ans $= \lfloor n/3 \rfloor$

Art Gallery Problem – Triangulation

Theorem 3.1 *Every simple polygon admits a triangulation, and any triangulation of a simple polygon with n vertices consists of exactly $n-2$ triangles.*

Theorem: (Art Gallery Theorem) For a simple polygon with n vertices, $\lfloor \frac{n}{3} \rfloor$ cameras are sufficient to have every point in the polygon visible from at least one of the cameras.